

BSP Release Note

for Windows[®] CE 5.0

MMSP[™] 2

Release Note

MDMP2520FBSPRN-WC50

Ver. 5.4.2 beta / August 3,2005

MP2520F

Platform Builder for Microsoft Windows[®] CE 5.0

for MMSP 2 Application Processor

MAGIC EYES

This document contains information on a product under development. MagicEyes reserves the right to change or discontinue this product without notice. © MagicEyes Digital Co.ltd 2004. All rights reserved.

Important Notice

Information in this document is provided solely to enable system and software implementers to use MagicEyes products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits or integrated circuits based on the information in this document.

MagicEyes reserves the right to make changes without further notice to any products herein. MagicEyes makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does MagicEyes assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters which may be provided in MagicEyes data sheets and/or specifications can and do vary in different applications and actual performance may vary over time. All operating parameters, including "Typicals" must be validated for each customer application by customer's technical experts. MagicEyes does not convey any license under its patent rights nor the rights of others. MagicEyes products are not designed, intended, or authorized for use as components in systems intended for surgical implant into the body, or other applications intended to support or sustain life, or for any other application in which the failure of the MagicEyes product could create a situation where personal injury or death may occur. Should Buyer purchase or use MagicEyes products for any such unintended or unauthorized application, Buyer shall indemnify and hold MagicEyes and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if such claim alleges that MagicEyes was negligent regarding the design or manufacture of the part.

© MagicEyes Digital

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electric or mechanical, by photocopying, recording, or otherwise, without the prior written consent of MagicEyes Digital.

HOW TO REACH US:

MagicEyes Digital co., Ltd.

463-824 #501, A-ONE Plaza, 269-6 Seohyeon-Dong, Bundang-Gu
Seongnam-City, Gyeonggi-Do, Korea
TEL: 82-31-788-0300
FAX:82-31-707-5765

HOME PAGE:

<http://www.mesdigital.com>

NOTICE

MAGICEYES INFORMS THAT THIS CODE AND INFORMATION IS PROVIDED "AS IS" BASE AND WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE IMPLIED WARRANTIES OF MERCHANTABILITY AND/OR FITNESS FOR A PARTICULAR PURPOSE.

Table of Contents

1.	Release Baseline.....	5
1.1.	Software Stack.....	5
1.2.	Software Modules	6
2.	Installation Attention	15
3.	New Components.....	16
4.	Modified Components	16
5.	Known Problems.....	17
5.1.	USB 2.0 Mass Storage	17
5.2.	MESPlayer	17
6.	CODEC Status	18
6.1.	MPEG4 Decoder.....	18
6.1.1.	Supported Video Stream Formats	18
6.1.2.	Performance	18
6.1.3.	Known Problems.....	18
6.2.	WMV Decoder.....	18
6.2.1.	Supported WMV Versions	18
6.2.2.	Performance	18
6.2.3.	Known Problems.....	18
7.	Release History	19

1. Release Baseline

1.1. Software Stack

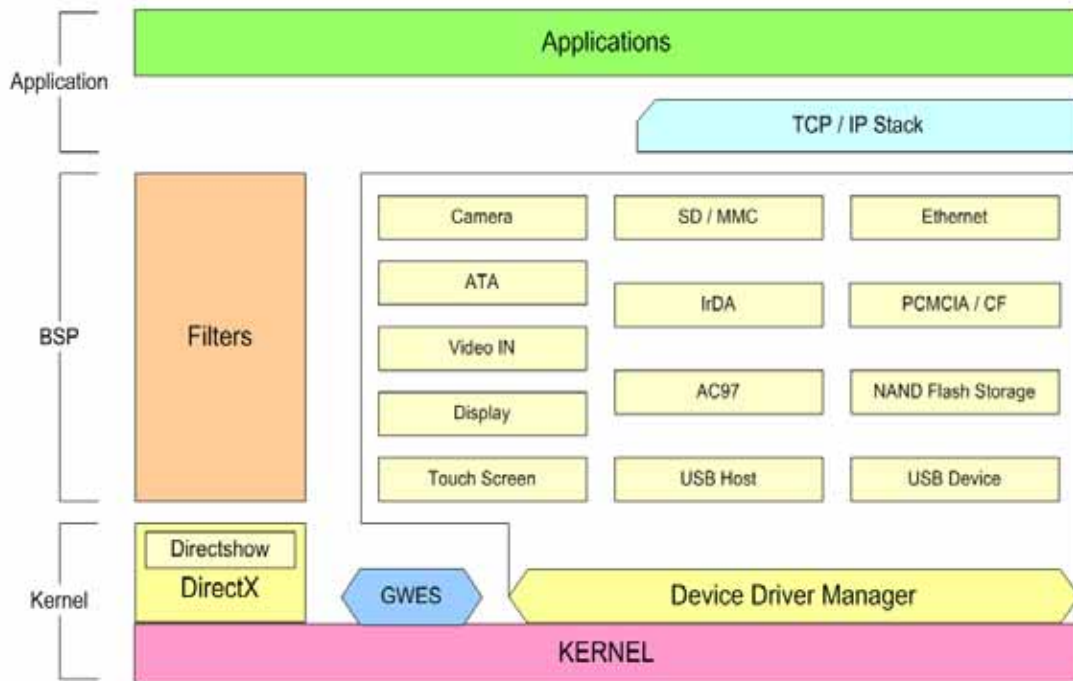


Figure 1. Software Stack

1.2. Software Modules

Basic

Kernel

The kernel is the core of the Windows CE operating system (OS). The kernel provides the base OS functionality for any Windows CE-based device. This functionality includes process, thread, and memory management. The kernel also provides some file management functionality. The kernel services enable applications to use this core functionality. MMSP2 BSP includes some kernel functions to run OS on MMSP2 based system.

Features	● OS initialization code ● interrupt handling of principal peripheral ● Power suspend/ wake-up ● Quick rebooting (boot from valid OS image in RAM without loading process) ● Remote Kernel Debugging with KITL interface
Not Supported Features	

Ethernet Bootloader

Ethernet bootloader receives Windows CE OS image (nk.bin) from platform builder through Ethernet link and load it to RAM of the MMSP2 based system. It can also store received image to the NAND flash of the system. If valid OS image is stored in NAND flash, Ethernet bootloader can load OS image from NAND flash to RAM.

Features	● Load image from Windows CE platform builder through Ethernet link ● Load image from the image which is stored in NAND flash. ● Quick booting mode support (To use this mode, OS image should be built with quick boot mode option.)
Not Supported Features	

Device Drivers

Display Driver

DirectDraw Display Driver

Features	● 8 / 16 / 24 bpp ● Output mode ■ LCD 640x480x60, 800x600x60, 800x480x60 ■ Conexant CX25874 ARGB : 640x480x60, 800x600x60 Composite/S-Video : 720x480x30 Component : 720x480x60 ■ Silicon Image SIL164 DVI 800x600x60 ● Hardware Overlay ■ 2 independent hardware overlay ■ YUV 4:2:2 format support ■ Fine or coarse scale up/down ■ Color control : brightness, contrast, hue, saturation ● Gamma correction ● Hardware cursor ● Bit-blit hardware acceleration ● OSD layer ● Sub-picture Layer ● Dynamic Rotation & Output Mode Change (GUI resolution is fixed.)
Sample Application	● APPS\COLORCTL ■ Sample application for color controls of overlay ● APPS\OSDDemo ■ Sample application for OSD layer ● APPS\SPUDemo ■ Sample application for SPU layer ● APPS\SetMode ■ Sample application for Dynamic Rotation & Output Mode Change. ● APPS\TestRGB ■ Sample application for controls of RGB Layer.

Omnivision OV9640 CMOS Sensor Driver

A device driver to support 1.3 mega pixel CMOS camera.

Feature	● Supports the OV9640 camera chip ● Get image data from camera and transfer to ISP ● Supports BayerRGB/CCIR656/CCIR656Like ● Supports VGA/QVGA/CIF/QCIF resolutions. ● Supports H/V mirrors. ● Supports to adjust a frame rate.
Sample Application	● FILTER\ISPSrc\MESCS ● APPS\CamView

Philips SAA7113 VideoDecoder Driver

A device driver to support a video decoder IC

Feature	● Supports the SAA7713 video decoder chip ● Get image data from external device and transfer it to ISP ● Supports S-Video/Composite inputs. ● Supports the TSC6/PTI5NF Tuner
---------	---

Sample Application	● FILTER\ISPSource\MESVIS ● APPS\VideoIn
--------------------	---

Ti TVP5150A VideoDecoder Driver

A device driver to support a video decoder IC

Feature	● Supports the TVP5150A video decoder chip ● Get image data from external device and transfer it to ISP ● Supports S-Video/Composite inputs.
Sample Application	● FILTER\ISPSource\MESVIS ● APPS\VideoIn

MMSP2ISP Driver

A device driver to support ISP & SCALER module of MMSP2

Feature	● Save image data from external device to memory ● Supports interfaces to set timing parameters ● Supports YUV 4:2:2 and 2D YUV 4:2:0 format
Future release	● Provides information for AF/AE/AWB control ● Provides interfaces for color controls
Sample Application	● FILTER\ISPSource\MESCS ● FILTER\ISPSource\MESVIS ● APPS\CamView ● APPS\VideoIn

I2C Driver

A device driver to support I2C interface

Feature	● Use GPIO module to make I2C signals instead of I2C module in the MMSP2
Future release	● Supports the Hardware I2C in the MMSP2

AC97 Driver

AC97 Audio Driver to play wave audio files.

Feature	● Supports REALTEK six channel AC97 audio codec ALC650 ● Compatible with AC97 2.2
---------	--

ATA Driver

A storage device driver that manages read/write operation to ATA hard disk.

Feature	● ATA Hard Disk Driver ● Support PIO / UDMA mode Default mode is PIO. UDMA mode conflicts with with MPEG H/W dec UDMA mode can be used only for file transfer
Not Supported	● ATAPI is not supported.
Features	

Limitations	- Maximum partition size is limited to 136GB. - This ATA disk driver can recognize partitions in ATA HDD that is created by desktop Windows, but mounting partition created with this driver on desktop Windows can be failed. - If a HDD has multiple partitions, mounting some partition can be failed. In this case, only primary partition will be mounted.
-------------	---

SD/MMC Storage Driver

A storage device driver that manages read/write operation to SD/MMC memory card.

Features	- SD memory card can be mounted as a storage device whenever the system runs (hot-swap of the card is supported) - Supports SD and High-speed MMC memory card.
Limitations	Some MMC card does not work.

NAND Flash Storage Driver

A storage device driver that maps part of NAND flash memory as a storage device.

Feature	Can use NAND flash memory space as a storage device except of domain that stores OS image.
Limitations	- support only 8-bit bus and size of <= 128MB - The size of NAND flash memory used for storage is fixed. - One bank with 32MB(8-bit bus, 3-addr type) NAND is tested.

USB Host Driver

A driver that manages USB devices connected to MMSP2 system. The universal serial bus (USB) is an external bus architecture for connecting a USB-compatible peripheral device such as a mouse or storage device to a host computer. USB is a communication protocol that supports serial data transfers between a host computer and a USB-compatible peripheral device.

Feature	- MMSP2 System works as an USB Host when USB Devices are connected. - Recognize mouse, keyboard, and storage device through class drivers supported by Windows CE.
---------	---

USB 1.1 Function Driver

A driver that makes MMSP2 system work as a USB device.

Feature	• USB 1.1 device controller on MMSP2 is used. • MMSP2 System works as USB device when a USB host is connected • ActiveSync link is possible with USB connection
---------	---

USB 2.0 Function Driver (Mass Storage Interface)

A driver that makes MMSP2 system work as an external mass storage when connected to a USB host.

Feature	• NET2272 USB 2.0 device controller is used. • MMSP2 System works as USB external storage device when USB host is connected
---------	--

CS8900 Ethernet Driver

A driver that facilitates communication between CS8900 Ethernet adapter card and network protocol stack provided with Windows CE operating system.

Feature	• Connect Local Area Network through Ethernet connection • Network features supported by Windows CE can be used. (Web browser, FTP server, etc.)
---------	---

PCMCIA / CF Driver

A driver that facilitates access service to PCMCIA/CF memory.

Feature	• Support PCMCIA / CF storage device. (CF/PCMCIA ATA storage driver that runs on this driver is required – Provided by Microsoft)
---------	---

Remote Controller Driver

A driver that receives input from IR remote controller and generate appropriate event.

Feature	• Sample driver for IR remote controller
Limitations	• This is not a general-purpose remote controller driver. You can use the source code of this driver for reference only.

Video Codecs

MPEG4 Decoder

MPEG4 Decoder is able to decode various types of MPEG4, such as DivX and XviD.
This CODEC supports only decoding. System layer is not included.

Feature	• DirectShow transform filter • Supported formats : DivX 3.11, DivX 4.x, DivX 5.x (SP and ASP), XviD Max. resolution : D1 (720x480) Max. bitrate : 3.5Mbps Max. frames per second : 30fps • Output format : 2D YUV
Performance	• See MPEG4 Decoder performance chart
Limitations	• Do not support 4MV + QPEL contents. • Do not support GMC which has 2 or more warping points. • In some DivX 3.11 contents you may see defections

WMV Decoder

WMV Decoder is able to decode WMV7, 8 and 9 contents
This CODEC supports only decoding. System layer is not included.

Feature	• DirectShow transform filter • Supported formats : WMV7, WMV8, WMV9 Max. resolution : QVGA (320x240) Max. bitrate : 500kbps Max. frames per second : 30fps • Output format : 2D YUV
Performance	• QVGA, 500Kbps, 30frames per second
Limitations	

Audio Codecs

Microsoft MP3 Decoder

MP3 Decoder is able to decode MP3 audio stream.

Feature	• MP3 Audio Decoder
---------	---

MES MP3 Decoder

MES MP3 Decoder evaluation version is included in BSP.

Feature	• MES MP3 Audio Decoder Eval.
---------	---

WMA Decoder

WMA Decoder is able to decode Windows Media Audio stream.

Feature	• Windows Media Audio Decoder
---------	---

Audio Capture

Audio capture filter receives audio input data for encoding.

Feature	• Audio Capture Source Filter • Capture from the microphone input • Fixed sample rate : 22050Hz
---------	---

Media File Formats

AVI

Feature	• Microsoft AVI File Format
---------	---

MPG

Feature	• MPEG1 File Format
---------	---

WMV

Feature	• WMV File Format
---------	---

ASF

Feature	• ASF File Format
---------	---

Other Media Components

DST AVI DEMUX

Feature	- Developed by DST Corporation. (Tel. +82-31-710-8800) - Demux Microsoft AVI File into audio and video stream
---------	--

DST AVI MUX

Feature	- Developed by DST Corporation. (Tel. +82-31-710-8800) - Audio and Video mux filter
---------	--

DST AVI Writer

Feature	- Developed by DST Corporation. (Tel. +82-31-710-8800) - Writes AVI multiplexed stream into a file
---------	---

Sample Applications

MESPlayer

MES Player is a simple media player for MPEG4 decoding and encoding functions.

Feature	- Play MPEG4 media file by using MPEG4 video decoder filter - Encode camera input to MPEG4 media file.
---------	---

OSImageUpdate

OSImageUpdate is a simple application program that replaces OS image in NAND flash with new image.

Feature	- Replace OS image with nk.nb0 file in ATA Drive. - Supports OS image file of ROM image format (nk.nb0) - Source code is in APPS\OSImageUpdate folder.
---------	---

Development Environment

TCP / IP Debugging

Remote debugging tools provided in development environment such as eMbedded Visual C++ or Windows CE Platform Builder can be connected to MMSP2 system through TCP/IP networking.

Feature	● Supports remote debugging of applications that runs on MMSP2 system through TCP/IP network (with eMbedded C++) ● Remote debugging tools are available. (ex. Remote process viewer)
---------	--

2. Installation Attention

- To avoid conflict, delete previous BSP before install current BSP.
- Import new .cec file into Platform Builder's **Manage Catalog Features** after delete previous .cec.
- Refer "**BSP Build Guide For MP2520F**" chapter of **MP2520F Development Kit User's Manual** document for information about making new platform and adding necessary components.
- For default LCD output, set DIP switch on the board as:
 - "Off Off Off Off" for DTK 4th board

3. New Components

This is the first version of Windows CE 5.0 BSP.

4. Modified Components

This is the first version of Windows CE 5.0 BSP.

5. Known Problems

5.1. USB 2.0 Mass Storage

- To use HDD as mass storage you must remove other storages from device before boot-up.

5.2. MESPlayer

- If you open contents several times MESPlayer can be unstable.

6. CODEC Status

6.1. MPEG4 Decoder

6.1.1. Supported Video Stream Formats

DivX 3.11, DivX 4.x, DivX 5.x SP, DivX 5.x ASP, XviD

6.1.2. Performance

DivX 4.x, DivX 5.x, XviD : 720x480, 3.5Mbps, 30fps

DivX 3.11 : 640x352, 1.8Mbps, 24fps

(for maximum performance you should use MES MP3 decoder and MES Player)

6.1.3. Known Problems

- In some DivX 3.11 contents you may see defections
- Media files without audio may have some problems, irregular playback speed, unable to repeat, etc.

6.2. WMV Decoder

6.2.1. Supported WMV Versions

WMV7, WMV8, WMV9

6.2.2. Performance

QVGA (320x240), 500kbps, 30fps

6.2.3. Known Problems

none

7. Release History

This is the first release of Windows CE 5.0 BSP. For release history of Windows CE.NET 4.2 BSP, see “BSP Release Note for Windows CE.NET 4.2